Programming Assignment 1: Stochastic Bandits

Wireless Channel Allocation with Two Feedback Modes

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1 Task 0: Modeling

The environment consists of a base station serving two users (UE1 and UE2) utilizing $K = 5$ orthogonal channels. At each time step $t = 1, 2, ..., T$, the base station allocates exactly one distinct channel to each user, ensuring $j_1 \neq j_2$. The baseline SNR parameters are sampled as $s_{i,j} \sim Unif[0.8, 2.0]$ with seed(28) as per requirements.

1.1 Case A: Aggregate Feedback Only

- **Action Space (Arms):** An action $a_t = (j_{1,t}, j_{2,t})$ is a unique permutation of two distinct channels. This constitutes a combinatorial space of $P(5, 2) = 20$ discrete arms.
- **Available Information:** The learner exclusively observes the aggregate system sum-rate $r_t = R_1^t(j_{1,t}) + R_2^t(j_{2,t})$. The individual components $R_1^t(j_{1,t})$ and $R_2^t(j_{2,t})$ are not observed.
- **Learning Target:** The algorithm estimates the expected aggregate sum-rate per combinatorial arm, $\mathbb{E}[r_t(a)]$.

1.2 Case B: Feedback Per User

- **Action Space (Arms):** The valid combinatorial actions remain the 20 distinct channel permutations. However, the system fundamentally tracks $2 \times 5 = 10$ individual base arms (user-channel pairs).
- **Available Information:** The learner observes the individual decoupled rates $(R_1^t(j_{1,t}), R_2^t(j_{2,t}))$.
- **Learning Target:** The algorithm estimates the expected individual rate per user-channel component, $\mathbb{E}[R_i^t(j)]$, constructing independent confidence bounds.

2 Tasks 1 & 2: UCB implementation for Case A & Case B

Both Case A and Case B employ the Upper Confidence Bound (UCB) algorithm. The theoretical exploration bounds are strictly scaled by the maximum possible observable reward in the system to avoid normalizing the raw sum or individual rates. The instantaneous SNR is modeled as $s_{i,j}^t \sim Unif[s_{i,j} - \epsilon, s_{i,j} + \epsilon]$ where $\epsilon = 0.6$.

2.1 Case A Configuration

The agent maintains 20 independent empirical means for the combinatorial pairs. The UCB bound is scaled by the theoretical maximum sum-rate $R_{max} = 2 \times \log(1 + 10(2.6))$. Due to the aggregate feedback constraint, a noisy observation penalizes the entire channel pair equally.

2.2 Case B Configuration

The architecture transitions to a Combinatorial UCB utilizing semi-bandit feedback. The agent tracks 10 independent components (5 for UE1, 5 for UE2), updating their bounds individually. The action selection metric is the summation of the independent upper confidence bounds of the constituent channels, scaled by $R_{max_ind} = \log(1 + 10(2.6))$.

3 Task 3: Experimental comparison

Experiments evaluate the algorithms across horizons $T \in \{2000, 20000\}$. All results are averaged across 50 Monte Carlo iterations keeping the baseline SNR matrix fixed and resampling per-epoch noise.

3.1 Theoretical Expectations

Case B is expected to significantly outperform Case A. Case A suffers from a credit assignment problem inherent to aggregated rewards. Case B observes exact component performance, converging on the optimal estimates with lower sample complexity by updating the statistical confidence of multiple permutations simultaneously through single component observations.

3.2 Empirical Observations

The empirical data confirms the theoretical expectations. Case B exhibits a substantially lower cumulative regret trajectory. The per-round regret variance highlights that Case B achieves consistent exploitation rapidly, whereas Case A requires prolonged exploration of suboptimal arms to satisfy the wide confidence intervals of 20 high-variance permutations.

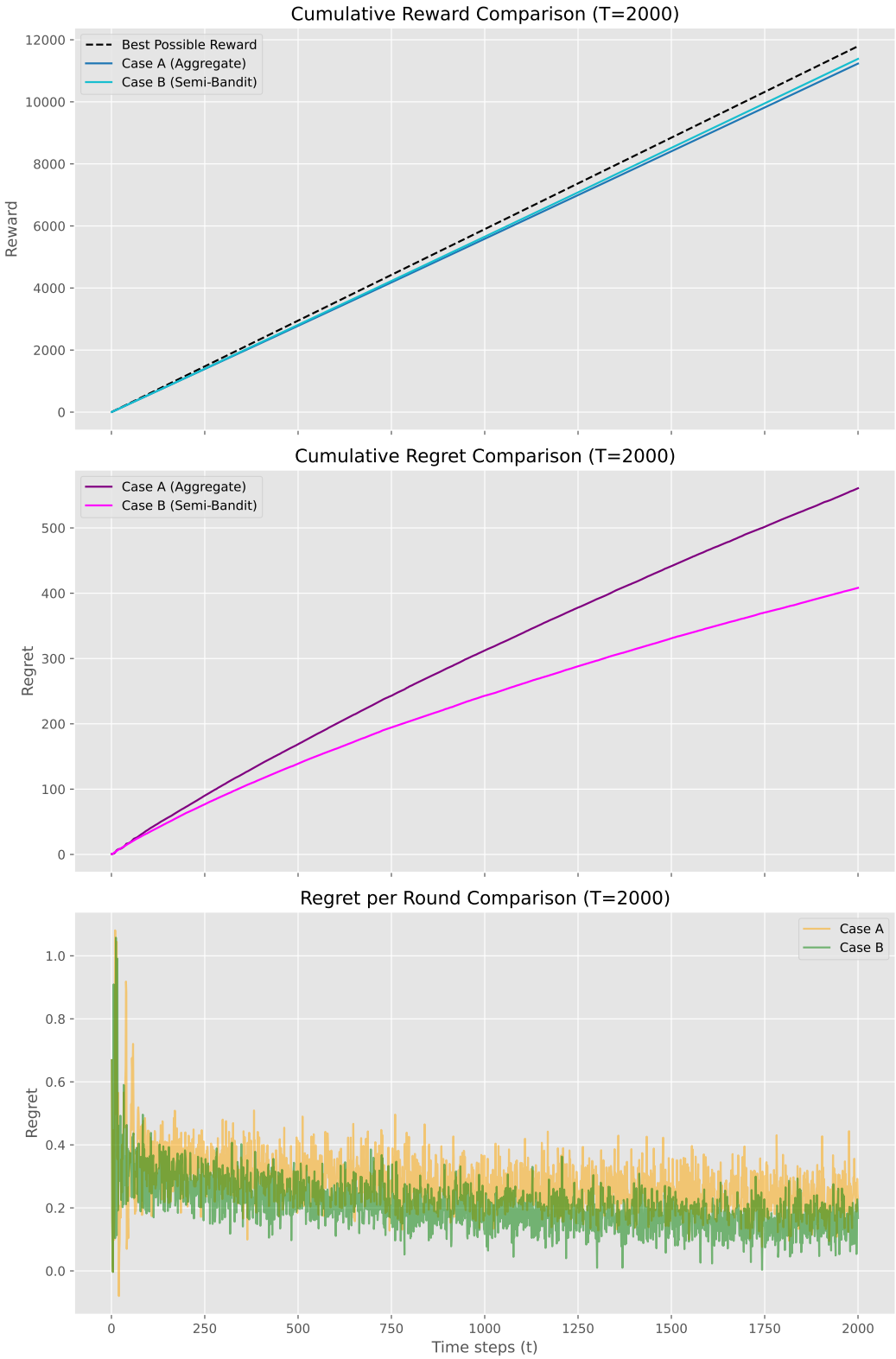

Figure 1: Experimental comparison of Case A and Case B across $T = 2000$.

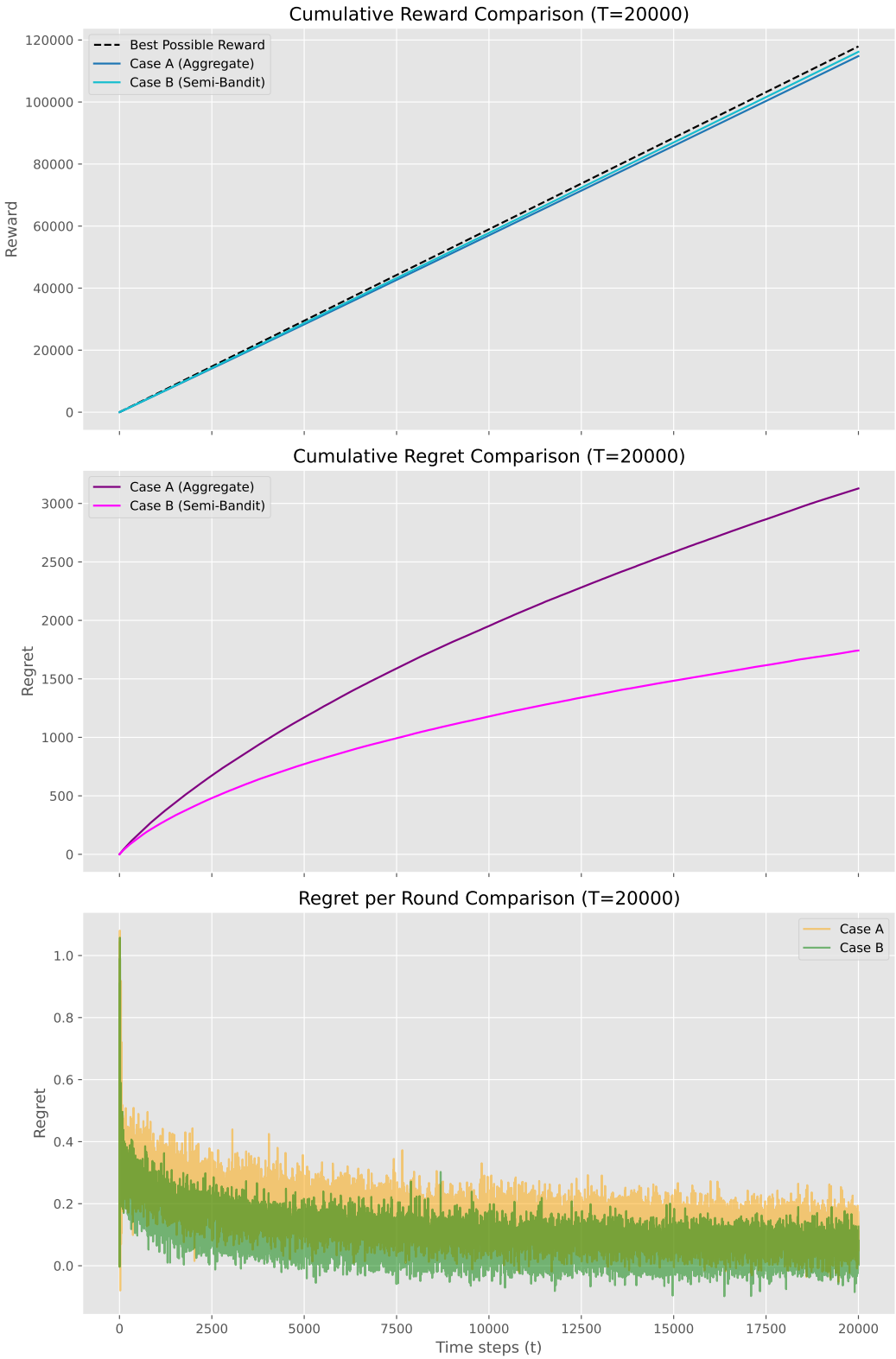

Figure 2: Experimental comparison of Case A and Case B across $T = 20000$.